Why pair the four subspaces?

For $A \in \mathbb{R}^{m \times n}$, each pair below shares an ambient space.

$$C(A^T), N(A) \subseteq \mathbb{R}^n,$$
$$C(A), N(A^T) \subseteq \mathbb{R}^m.$$

Their dimensions sum to n and m, respectively.

$$\dim C(A^T) + \dim N(A) = r + (n - r) = n,$$
$$\dim C(A) + \dim N(A^T) = r + (m - r) = m.$$

These pairs are orthogonal as well as dimensionally complementary. Every input decomposes uniquely into row-space and nullspace components. Every output decomposes into column-space and left-nullspace components.

Fix the ambient space before taking inner products. A vector in $\mathbb{R}^m$ and one in $\mathbb{R}^n$ cannot be paired by the ordinary dot product unless the dimensions and coordinate interpretation agree.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Why row space is orthogonal to nullspace

If $x \in N(A)$, then $Ax = 0$. Expanding by rows gives

$$\begin{bmatrix} r_1^T \\ \vdots \\ r_m^T \end{bmatrix} x = \begin{bmatrix} r_1^T x \\ \vdots \\ r_m^T x \end{bmatrix} = 0.$$

Thus x is orthogonal to each row. Linearity makes it orthogonal to every linear combination of rows.

$$x^T(c_1 r_1 + \cdots + c_m r_m) = 0.$$

Conversely, orthogonality to every row gives $Ax = 0$. This proves $N(A) = C(A^T)^\perp$. The spaces are orthogonal and their dimensions sum to n, so each is the other's orthogonal complement. Hence

$$C(A^T) = N(A)^\perp.$$

Replacing A by A^T gives $C(A) = N(A^T)^\perp$.

$U^\perp$ consists of all vectors orthogonal to every vector of U. Orthogonality plus the dimension sum establishes the full complement relation.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Orthogonality and complements

Write $U \perp W$ when $u^T w = 0$ for every $u \in U, w \in W$. If $z \in U \cap W$, then $z^T z = 0$, forcing $z = 0$. Therefore $U \cap W = \{0\}$. The converse is false.

In a finite-dimensional inner-product space, if

$$U \cap W = \{0\}, \qquad \dim U + \dim W = \dim V,$$

then $V = U \oplus W$, a direct sum. The subspaces can still meet at an oblique angle. An orthogonal direct sum also requires $U \perp W$.

For example, $U = \operatorname{span}\{(1,0)^T\}$ and $W = \operatorname{span}\{(1,1)^T\}$ intersect only at zero and their dimensions sum to 2. They are not perpendicular. Replacing W by $\operatorname{span}\{(0,1)^T\}$ gives an orthogonal complement.

The inner product determines the orthogonal complement. Complex spaces require conjugate transposes.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Dimensions and a decomposition example

Since $C(A^T) \oplus N(A) = \mathbb{R}^n$, every $x \in \mathbb{R}^n$ has the unique decomposition

$$x = x_{\text{row}} + x_{\text{null}}, \qquad x_{\text{row}}^T x_{\text{null}} = 0.$$

Take

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

whose rank is 2. Its two independent rows form a row-space basis, while the nullspace is spanned by

$$(-2, 1, -1)^T$$

Check directly that

$$(1, 2, 0)(-2, 1, -1)^T = 0, \qquad (0, 1, 1)(-2, 1, -1)^T = 0.$$

Thus $\mathbb{R}^3 = C(A^T) \oplus N(A)$. Since A has full row rank, $N(A^T) = \{0\}$ and $C(A) = \mathbb{R}^2$.

When a subspace is the whole ambient space, its orthogonal complement is the zero subspace. The resulting decomposition with a zero component is still unique.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Projection onto a line

For nonzero $a \in \mathbb{R}^n$, remove from b its component perpendicular to $L = \operatorname{span}\{a\}$. Write the projection as

$$p = \lambda a$$

and require $e = b - p$ to be perpendicular to the whole line.

$$a^T e = a^T (b - \lambda a) = 0.$$

Since $a^T a > 0$,

$$\lambda = \frac{a^T b}{a^T a}, \qquad p = a \frac{a^T b}{a^T a}.$$

The projection matrix is

$$P = \frac{a a^T}{a^T a}, \qquad p = Pb.$$

For $a = (1, 1, 1)^T$, $b = (2, 1, 4)^T$, we have $a^T b = 7$, $a^T a = 3$, and

$$p = \frac{7}{3}(1, 1, 1)^T, \quad e = b - p = \frac{1}{3}(-1, -4, 5)^T.$$

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Projection onto a line (continued)

Direct substitution gives

$$p + e = (2, 1, 4)^T = b, \qquad a^T e = \frac{1(-1) + 1(-4) + 1(5)}{3} = 0.$$

Hence $p^T e = 0$, and

$$\|e\|_2 = \frac{\sqrt{(-1)^2 + (-4)^2 + 5^2}}{3} = \frac{\sqrt{42}}{3}.$$

The same values satisfy the normal equation

$$a^T a \, \lambda = a^T b, \qquad 3 \cdot \frac{7}{3} = 7.$$

The diagram uses the orthonormal basis $f_1 = a/\sqrt{3}$, $f_2 = e/(\sqrt{42}/3)$ in $\operatorname{span}\{a, e\}$. Its two-dimensional coordinates are

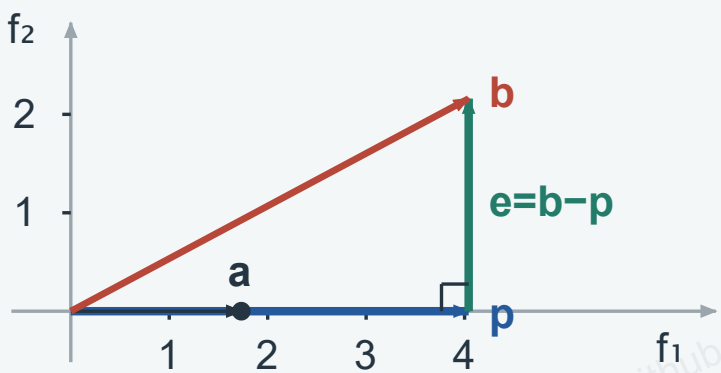
$$[a] = (\sqrt{3}, 0), \quad [p] = (7/\sqrt{3}, 0), \quad [e] = (0, \sqrt{42}/3), \quad [b] = (7/\sqrt{3}, \sqrt{42}/3).$$

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Projection onto a line (continued)

These coordinates give an isometry from that plane to $\mathbb{R}^2$, with equal scales on both axes. The right angle represents the original inner product. The short black vector a and longer blue vector $p = (7/3)a$ have distinct endpoints. The green residual is translated to run from p to b.

Exact projection: b=p+e in orthogonal coordinates



Orthonormal axes; equal scale

$f_1 = a/\sqrt{3} \cdot f_2 = e/(\sqrt{42}/3)$

$a = (\sqrt{3}, 0) \cdot p = (7/\sqrt{3}, 0)$

$e = (0, \sqrt{42}/3) \cdot b = p + e$

Original $a = (1,1,1)$, $b = (2,1,4)$

Orthogonality also controls maximal growth directions in the SVD. Here it first characterizes the minimum-residual projection.

Projection onto a subspace

Let $A \in \mathbb{R}^{m \times n}$ have independent columns, and let

$$S = C(A).$$

The projection of b onto S has residual $e = b - Ax$ orthogonal to every basis vector of S.

$$A^T(b - Ax) = 0.$$

This gives the normal equations

$$A^T Ax = A^T b.$$

With full column rank, $A^T A$ is invertible, so

$$p = A(A^T A)^{-1} A^T b.$$

and the projection matrix is

$$P = A(A^T A)^{-1} A^T.$$

The projector P is not A^{-1}, since A need not be square. Nor may $(A^T A)^{-1}$ be replaced by $A^{-1}(A^T)^{-1}$ unless A is square and invertible.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Projection onto a subspace (continued)

Projection onto a line is the case $n = 1$. Independent columns make $A^T A$ positive definite and invertible because $z^T A^T A z = \|Az\|^2$.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Why column independence is necessary

For every $z \in \mathbb{R}^n$,

$$z^T A^T A z = (Az)^T (Az) = \|Az\|^2 \geq 0.$$

Independent columns imply $Az \neq 0$ for nonzero z. The quadratic form is then strictly positive, so $A^T A$ is positive definite and invertible.

Conversely, dependent columns give a nonzero z with $Az = 0$. Then

$$A^T A z = A^T 0 = 0,$$

so $A^T A$ is singular. In complex spaces replace transpose by conjugate transpose.

Rank deficiency does not destroy uniqueness of the projected vector. It creates redundant parameters, so multiple coefficient vectors can produce the same projection.

Independent columns are necessary and sufficient for invertibility of $A^T A$. Full row rank does not suffice when $m < n$.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Algebraic properties of a projector

For full column rank A, let

$$P = A(A^T A)^{-1} A^T.$$

The matrix is symmetric and idempotent.

$$P^T = A((A^T A)^{-1})^T A^T = A(A^T A)^{-1} A^T = P,$$
$$P^2 = A(A^T A)^{-1}(A^T A)(A^T A)^{-1} A^T = P.$$

Idempotence means that projecting twice changes nothing further. Symmetry distinguishes an orthogonal projector from a general oblique projector. The complementary projector

$$I - P$$

projects onto $C(A)^\perp = N(A^T)$, and $P(I - P) = 0$.

Idempotence alone does not imply orthogonality. Symmetry is essential in the real Euclidean setting.

The nearest point and its error

Let $p = Pb$ and $e = (I - P)b$. Then

$$b = p + e, \qquad p^T e = 0.$$

Pythagoras gives

$$\|b\|^2 = \|p\|^2 + \|e\|^2.$$

For any $q \in C(A)$, the difference $p - q$ lies in $C(A)$, while e is perpendicular to that space. Therefore

$$\|b - q\|^2 = \|e + (p - q)\|^2 = \|e\|^2 + \|p - q\|^2.$$

The right side is minimized uniquely at $q = p$. This proves the nearest-point property without calculus.

The cross term vanishes by orthogonality. In a complex inner-product space the corresponding norm expansion uses conjugate transposes and real parts.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Least squares

Even when $Ax = b$ is inconsistent, choose

$$x^\star \in \arg\min_x \|b - Ax\|_2^2.$$

The fitted vector $p = Ax^\star$ is the unique projection onto $C(A)$. Full column rank makes the parameter $x^\star$ unique as well. Rank deficiency gives infinitely many minimizers differing by nullspace vectors.

Minimizing the norm or its square gives the same minimizers because squaring is strictly increasing on $[0, \infty)$. The square avoids a square root and gives a linear gradient equation.

The objective minimizes a scalar residual norm, not the vector residual under an unspecified ordering. In general the fitted vector differs from b.

Least squares minimizes the sum of squared coordinate errors. The optimal residual is perpendicular to the attainable subspace; it need not vanish.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Deriving the normal equations

Define

$$E(x) = (b - Ax)^T (b - Ax).$$

For any displacement h, expand the square.

$$\begin{aligned} E(x + h) - E(x) = & -2h^T A^T (b - Ax) \\ & + h^T A^T A h. \end{aligned}$$

The gradient is therefore

$$\nabla_x E = -2A^T (b - Ax).$$

At an unconstrained minimum $x^\star$, the first-order change vanishes in every direction h, so

$$A^T (b - Ax) = 0 \iff A^T A x = A^T b.$$

With full column rank, $h^T A^T A h = \|Ah\|^2 > 0$ for every nonzero h. The quadratic remainder is strictly positive, proving the unique global minimum. Both sides of the normal equations have shape $n \times 1$.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Deriving the normal equations (continued)

For dependent columns, the normal equations may have multiple solutions. If x, z both solve them, let $q = x - z$. Subtracting gives $A^T Aq = 0$. Multiplying by q^T,

$$0 = q^T A^T Aq = \|Aq\|^2.$$

so $Ax = Az$. The parameter vectors can differ while the projection point remains unique.

A^T is not the projection matrix. The condition $A^T e = 0$ tests orthogonality to every column, equivalently $e \in N(A^T)$.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Fitting a straight line

Fit $y = c + mt$ to

$$(t, y) = (0, 1), (1, 2), (2, 2).$$

The design matrix and vectors are

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} c \\ m \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

Compute

$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Solving gives

$$m = \frac{1}{2}, \qquad c = \frac{7}{6}.$$

The fitted vector is

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Fitting a straight line (continued)

$$p = \left(\frac{7}{6}, \frac{5}{3}, \frac{13}{6}\right)^T.$$

The residual is

$$e = b - p = \left(-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}\right)^T.$$

Its entries sum to zero, and $A^T e = 0$. The sum condition comes from the constant column. The full normal equation also checks orthogonality to the trend column t.

A zero sum of residuals alone does not prove least-squares optimality. Test the residual against every design column.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

rank-deficient least squares

Take

$$A = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

The two columns of A are identical, so its rank is 1. Projection onto their line gives

$$p = \frac{a^T b}{a^T a} a = \frac{7}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7/5 \\ 14/5 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} x = \begin{bmatrix} 7 \\ 7 \end{bmatrix},$$

and impose only

$$x_1 + x_2 = \frac{7}{5}.$$

Every parameter vector satisfying this relation produces the same $Ax = p$ and minimizes the error. The residual

Lin Cing https://github.com/xinhi6716/computer-science-self-study

rank-deficient least squares (continued)

$$e = \left(-\frac{2}{5}, \frac{1}{5}\right)^T$$

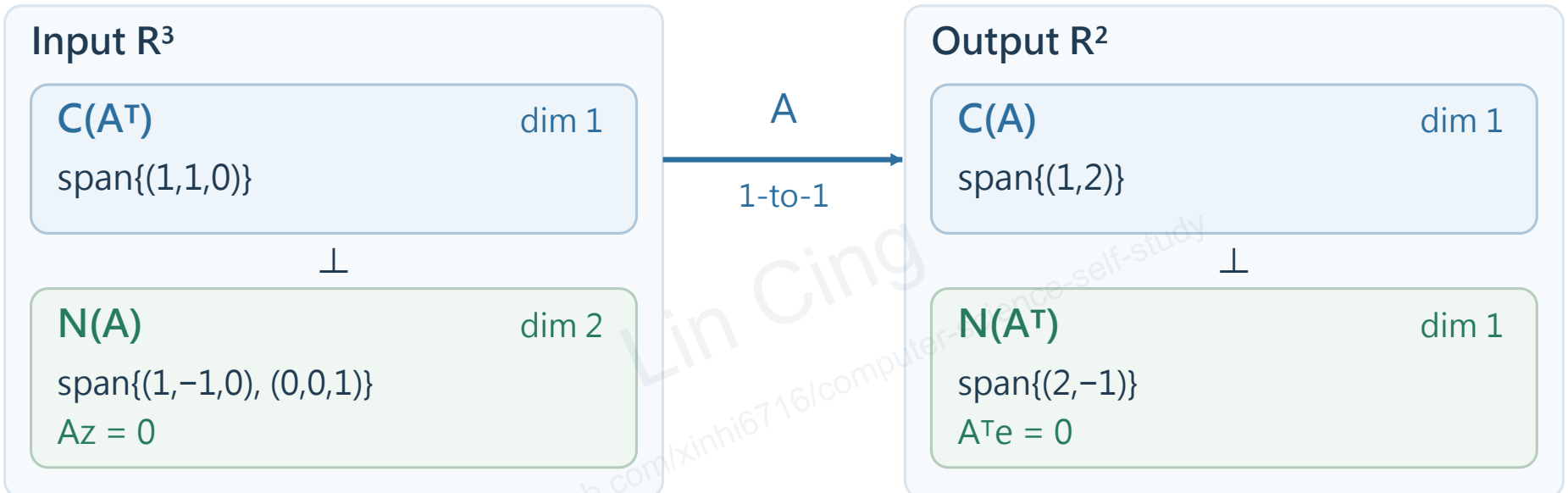
is orthogonal to $(1, 2)^T$. Thus projection uniqueness and parameter uniqueness are separate properties.

For deficient rank, use the pseudoinverse minimum-norm solution or explicit regularization; do not invert $A^T A$.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Four subspaces of one rectangular matrix

Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \end{bmatrix}$, of rank one. Inputs are in $\mathbb{R}^3$; outputs are in $\mathbb{R}^2$.



Boxes name subspaces and dimensions; their positions are not vector coordinates.

A maps row space one-to-one onto column space and sends N(A) to zero. Orthogonal residuals lie in $N(A^T)$.

A unique projection need not give unique parameters

For the same A, take $b = (3, 1)^T$. With $c = (1, 2)^T$, the column space is its span, so

$$p = \frac{c^T b}{c^T c} c = c = (1, 2)^T, \qquad e = b - p = (2, -1)^T.$$

Since $c^T e = 0$ and $p - Ax$ always lies in the column space,

$$\|b - Ax\|^2 = \|e\|^2 + \|p - Ax\|^2 = 5 + \|p - Ax\|^2.$$

The minimum residual norm is $\sqrt{5}$, attained exactly when Ax=p. This requires only $x_1 + x_2 = 1$, with x_3 free. All least-squares solutions are

$$x = \begin{pmatrix} 1/2 + t \\ 1/2 - t \\ s \end{pmatrix}, \qquad t, s \in \mathbb{R}.$$

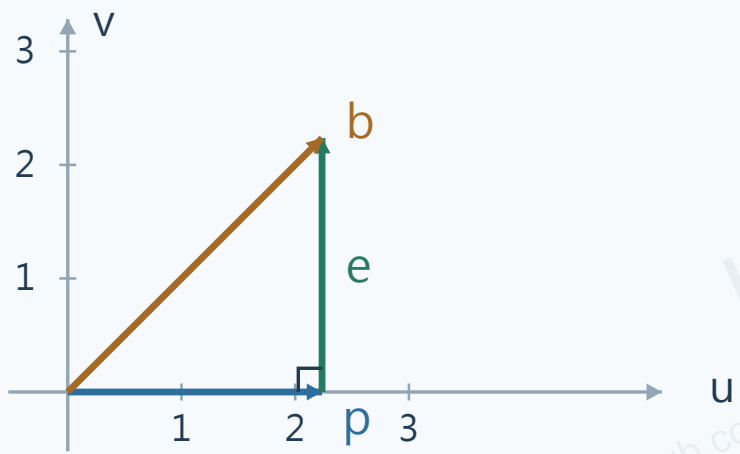
Lin Cing https://github.com/xinhi6716/computer-science-self-study

One minimum-norm solution among infinitely many solutions

Here $\hat{x} = (1/2, 1/2, 0)^T$ lies in the row space and $z = (t, -t, s)^T$ in the nullspace. Orthogonality gives

$$x = \hat{x} + z, \qquad \|x\|^2 = \tfrac{1}{2} + 2t^2 + s^2.$$

Output: orthonormal coordinates



[p]=(√5,0), [e]=(0,√5), [b]=(√5,√5)

All least-squares parameters

x = (1/2+t, 1/2−t, s)

Ax = p = (1,2)

||x||² = 1/2 + 2t² + s²

t=s=0: x̂ = (1/2,1/2,0)

p is unique; x need not be.

The minimum-norm parameter x̂ is unique.

The plot uses orthonormal axes $u = (1, 2)^T/\sqrt{5}$, $v = (2, -1)^T/\sqrt{5}$. The norm is minimized only at t=s=0. The pseudoinverse in Chapter 7 selects this same $\hat{x}$.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Orthonormal bases

Vectors $q_1, \ldots, q_k$ satisfying

$$q_i^T q_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases}$$

are orthonormal. Placing them in $Q \in \mathbb{R}^{m \times k}$ gives

$$Q^T Q = I_k.$$

If Q is square, these independent columns form an orthonormal basis of the whole space. Then $Q^{-1} = Q^T$, and Q is an orthogonal matrix.

If Q is rectangular, $Q^T Q = I_k$ gives full column rank, not $QQ^T = I_m$. The two identities have different sizes and different meanings.

Despite the historical name, an orthogonal matrix has orthonormal columns. A rectangular matrix with orthonormal columns has no two-sided inverse.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Gram–Schmidt construction

For independent vectors $a_1, \ldots, a_n$, begin by normalizing

$$q_1 = \frac{a_1}{\|a_1\|}.$$

For a_j, subtract its projections onto the existing directions.

$$z_j = a_j - \sum_{i=1}^{j-1} (q_i^T a_j) q_i,$$

Then normalize

$$q_j = \frac{z_j}{\|z_j\|}.$$

Independence guarantees $z_j \neq 0$. Otherwise a_j would be in the span of its predecessors. For $i < j$, expand $q_i^T z_j$ to obtain

$$q_i^T z_j = q_i^T a_j - \sum_{\ell=1}^{j-1} (q_\ell^T a_j)(q_i^T q_\ell) = q_i^T a_j - q_i^T a_j = 0.$$

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Gram–Schmidt construction (continued)

Inductively, the first $j-1$ vectors q are orthonormal and span $\operatorname{span}\{a_1,\dots,a_{j-1}\}$. Since a_j equals z_j plus a combination of the earlier directions, replacing z_j by its normalization q_j preserves the expanded span. At every step,

$$\operatorname{span}\{q_1,\dots,q_j\} = \operatorname{span}\{a_1,\dots,a_j\}.$$

For $a_1 = (1,1,0)^T, a_2 = (1,0,1)^T$,

$$q_1 = \frac{1}{\sqrt{2}}(1,1,0)^T,$$

$$z_2 = a_2 - (q_1^T a_2) q_1 = \left(\frac{1}{2}, -\frac{1}{2}, 1\right)^T,$$

$$q_2 = \frac{1}{\sqrt{6}}(1,-1,2)^T.$$

In floating point, a small $\|z_j\|$ signals near dependence, and division by it amplifies error. Chapter 11 develops more stable alternatives.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

QR and least squares

Collect the Gram–Schmidt expansions into

$$A = QR,$$

where Q has orthonormal columns and R is upper triangular. The coefficients $R_{ij} = q_i^T a_j$ for $i < j$, and $R_{jj} = \|z_j\|$, give $A = QR$ by column multiplication. In the preceding example,

$$R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{bmatrix}.$$

For full column rank, least squares reduces to

$$R^T Q^T Q R x = R^T Q^T b \Longrightarrow Rx = Q^T b.$$

Solve the invertible upper-triangular system in R. This avoids forming $A^T A$ and is more stable. For a rectangular A, the reduced Q is $m \times n$.

Do not cancel Q^T arbitrarily. In the normal-equation derivation one can cancel the invertible factor R^T, since full column rank guarantees invertible R.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

QR and least squares (continued)

Classical Gram–Schmidt is convenient for proofs. Numerical implementations usually use modified Gram–Schmidt or Householder QR, with the same exact mathematical factorization structure.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

A complete QR least-squares example

Use the preceding

$$Q = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{2} & -1/\sqrt{6} \\ 0 & 2/\sqrt{6} \end{bmatrix}, \quad R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{bmatrix}.$$

and $b = (1, 2, 0)^T$. First compute

$$Q^T b = \left(\frac{3}{\sqrt{2}}, -\frac{1}{\sqrt{6}} \right)^T.$$

Solve $Rx = Q^T b$. The second equation gives

$$\sqrt{\frac{3}{2}}\, x_2 = -\frac{1}{\sqrt{6}} \quad \Longrightarrow \quad x_2 = -\frac{1}{3}.$$

The first equation is

$$\sqrt{2}\, x_1 + \frac{1}{\sqrt{2}} \left(-\frac{1}{3} \right) = \frac{3}{\sqrt{2}},$$

Lin Cing https://github.com/xinhi6716/computer-science-self-study

A complete QR least-squares example (continued)

so $x_1 = 5/3$. The projection is

$$p = Ax = \left(\frac{4}{3}, \frac{5}{3}, -\frac{1}{3}\right)^T,$$

and the residual is

$$e = b - p = \frac{1}{3}(-1, 1, 1)^T.$$

Check directly that

$$a_1^T e = 0, \qquad a_2^T e = 0,$$

Thus $A^T e = 0$ and $\|e\|^2 = 1/3$. The nonzero residual makes the normal-equation orthogonality condition observable.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Chapter check

Distinguish the following concepts.

- $U \perp W$, $U \cap W = \{0\}$, and $V = U \oplus W$.
- Line projection, subspace projection, and a projection matrix.
- Unique least-squares parameters at full column rank and nonunique minimizers at deficient rank.
- $P^2 = P = P^T$ and the complementary projector $I - P$.
- $Q^T Q = I$ for orthonormal columns and $QQ^T = I$ for a square orthogonal matrix in this setting.
- Gram-Schmidt, $A = QR$, and preservation of the column span.

Check projection and least-squares calculations with $A^T e = 0$. A small residual alone does not prove the required orthogonality.

Chapter 5 connects determinants to invertibility and volume. Exact invertibility alone does not guarantee a numerically well-conditioned solve.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study

Lin Cing
https://github.com/xinhi6716/computer-science-self-study